

Unit 12, Lesson 6: Lengths of Secants, Tangents, Segments, and Chords in Circles



Benchmark

MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle.



Learning Target

- ☐ I can solve mathematical and real-world problems involving the lengths of secants, tangents, segments, or chords of a circle.



12.6.1 Warm-Up: Critical Thinking

1. Explain how math could be considered a language used for communication.
2. Suppose you were taking Geometry in another country. How do you think the problems would differ?



12.6.2 Activity: Line Segments of Circles

Match each image, type of segment, and answer to the description of circles. Record your answers in the table.

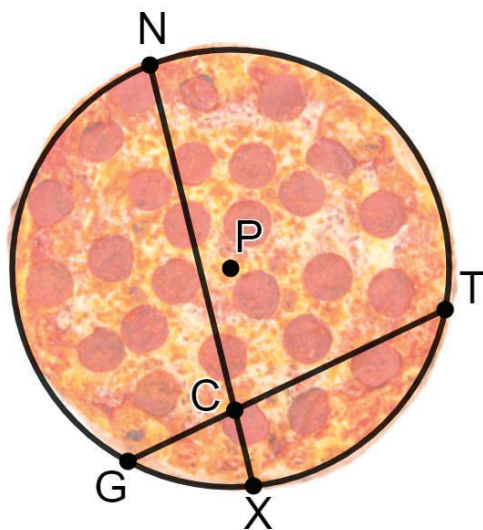
Circle	Work	Cards' Letter
Circle K has points G, N, D , and E on the circle, where $GM = 2x + 2$, $MD = 14$, $EM = 12$, and $MN = 2x + 5$		Image: _____ Segment: _____ Answer: _____
Circle H has points C, T , and B on the circle, where $BR = 15$, $RT = 9$, and $CT = x$		Image: _____ Segment: _____ Answer: _____
Circle R has points T, S, V , and W on the circle, where $TS = x - 8$, $SP = 4$, $VP = 3$, and $VW = x - 5$		Image: _____ Segment: _____ Answer: _____

Circle	Work	Cards' Letter
Circle F has points E and B on the circle, where $EY = 7x - 3$ and $BY = 5x + 1$		Image: _____ Segment: _____ Answer: _____
Circle V has point B on the circle, where $VB = 4$, $BA = 8$ and $VA = x$		Image: _____ Segment: _____ Answer: _____
Circle A has points G, R, H , and S on the circle, where $MA = WA$, $GR = 35$, and $SH = 6x - 7$		Image: _____ Segment: _____ Answer: _____



12.6.3 Try It: Solving for Length

1. When Doug's pizza was delivered, he realized the pizza was not cut evenly. The pizza, circle P , is cut by chords $\overline{NX}$ and $\overline{GT}$.



Given that $NC = (5x - 6)$ inches (in.), $CX = 2$ in., $GC = 3$ in., and $CT = 2x$ in., what is the length of $\overline{NX}$?

2. A belt guard is used on stationary bikes to cover some of the inner mechanics. Circle S is shown, where points H and T are on the circle. Tangents $\overline{HG}$ and $\overline{TG}$ intersect outside the circle at point G .



If $HG = 15$ and $TG = 4y - 9$, what is the value of y ?



12.6.4 Wrap-Up: Cool Down

1. Write **one question** you still have about solving mathematical and real-world problems involving the lengths of secants, tangents, segments, or chords of a circle.

Unit 12, Lesson 7: Deriving the Equation of a Circle



Benchmark

MA.912.GR.7.2 Given a mathematical or real-world context, derive and create the equation of a circle using key features.



Learning Targets

- ☐ I can use the Pythagorean Theorem to derive the equation of a circle.
- ☐ I can determine key features of a circle given the graph or equation.
- ☐ I can write the equation of a circle using key features.